CODEFORCES
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Codeforces Round 1133 (Div. 1, Div. 2) is coming soon! Register here.

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H. Coprime Graph

time limit per test: 2 seconds
memory limit per test: 256 megabytes

Forbidden Keywords for the Quiz: open, file, creat(, fstream, thread, process, system(, exec(

You are given an integer N. Construct an undirected graph with N nodes, where each node i is connected to all node j such that $gcd(i, j) = 1$ where $1 \leq i, j \leq N$ and $i \neq j$.

For example,for $N = 6$, the graph will be, $G = [[2, 3, 4, 5, 6], [1, 3, 5], [1, 2, 4, 5], [1, 3, 5], [1, 2, 3, 4, 6], [1, 5]]$.

Now, there will be Q queries. Each query consists of two integers X and K. For each query, you have to determine the $K - th$ smallest node connected to node X.

Input

The first line contains two integers N and Q $(1 \leq N \leq 2 \times 10^3, 1 \leq Q \leq 3 \times 10^5)$ — the number of vertices and the total number of queries.

The next Q lines contain two integers X and K $(1 \leq X \leq N, 1 \leq K \leq 10^6)$, representing a query.

Output

For each query, output the $K - th$ smallest node connected to node X. If there are fewer than K neighbors of X, then print -1.

Examples

input Copy

```
5 6
1 3
3 1
4 2
5 5
3 4
5 2
```

output Copy

```
4
1
3
-1
5
2
```

input Copy

```
2000 3
903 24
702 563
942 50
```

output Copy

CSE221 Assignment 04 Fall 2026
Contest is running
8 days
Contestant

→ Submit?

Language: Java 21 64bit

Choose file: Choose File No file chosen

Success!

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Submit

```
41
1829
149
```

input Copy

```
1 1
1 1
```

output Copy

```
-1
```

input Copy

```
2 1
2 1
```

output Copy

```
1
```

Note

Explanation of the First Sample (Let's go through the queries):

Query (1, 3): The neighbors of node 1 are [2, 3, 4, 5]. Sorted: [2, 3, 4, 5]. The 3rd smallest is 4. Output: 4.

Query (3, 1): The neighbors of node 3 are [1, 2, 4, 5]. Sorted: [1, 2, 4, 5]. The 1st smallest is 1. Output: 1.

Query (4, 2): The neighbors of node 4 are [1, 3, 5]. Sorted: [1, 3, 5]. The 2nd smallest is 3. Output: 3.

Query (5, 5): The neighbors of node 5 are [1, 2, 3, 4]. There are only 4 neighbors, so the 5th smallest does not exist. Output: -1.

Query (3, 4): The neighbors of node 3 are [1, 2, 4, 5]. Sorted: [1, 2, 4, 5]. The 4th smallest is 5. Output: 5.

Query (5, 2): The neighbors of node 5 are [1, 2, 3, 4]. Sorted: [1, 2, 3, 4]. The 2nd smallest is 2. Output: 2.

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